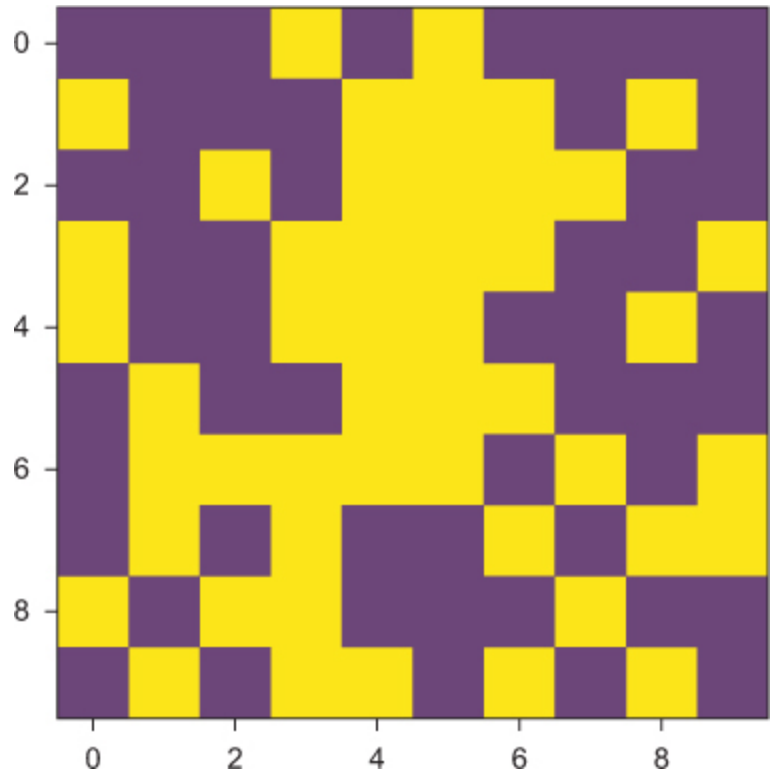
reduce the computational cost. Notice that we're only generating a single parameter vector; we're going to be using a single DQN to control all of the 100 agents, since they have the same optimal policy. We're creating two copies of the main grid for reasons that will be clear once we get to the training loop.

Figure 9.21. This is a randomly initialized 2D Ising model. Each grid square represents an electron. The light-colored grid squares represent electrons oriented with spin-up and the dark squares are spin-down.



For this example, we are going to add some of the complexities we left out of the 1D case, since this is a harder problem. We will be using an experience replay mechanism to store experiences and train on mini-batches of these experiences. This reduces the variance in the gradients and stabilizes the training. We will also use the proper target Q values, $r_{t+1} + \gamma * V(S_{t+1})$, so we need to calculate Q values twice per iteration: once to decide which action to take, and then again to get $V(S_{t+1})$. In listing 9.12 we jump into the main training loop of the 2D Ising model.

Listing 9.12. Mean field Q-learning: The main training loop